What Do We Mean by "Intelligence"?

Many abilities. One amazing mind.

Human intelligence is **not a single ability.** It includes **several different capabilities.**

Examples include:

Perception
Recognizing objects, sounds, faces, or situations.

Learning
Improving performance using experience.

Reasoning
Drawing conclusions from available information.

Problem-solving
Finding solutions to new situations.

Planning
Deciding which actions should be taken and in what order.

Language ability
Understanding and producing communication.

Adaptation
Changing behaviour when circumstances change.

Decision-making
Choosing among alternatives.



Human Intelligence
- Flexible
- General-purpose
- Adaptable
- Emotionally aware
- Learns from life experience

AI Systems
- Strong in specific tasks
- Learns from data
- No real emotions
- Often narrow in scope
- Still developing general intelligence

AI systems attempt to reproduce **some of these abilities computationally.**
However, an AI system may be **extremely capable in one area and weak in another.**

Different strengths. Shared potential.

Four Classical Views of Artificial Intelligence

Different questions, different perspectives, a richer understanding.

AI has traditionally been understood through **four major approaches.**

1 Thinking Humanly

Main Question:

Can a machine think in the same way humans think?

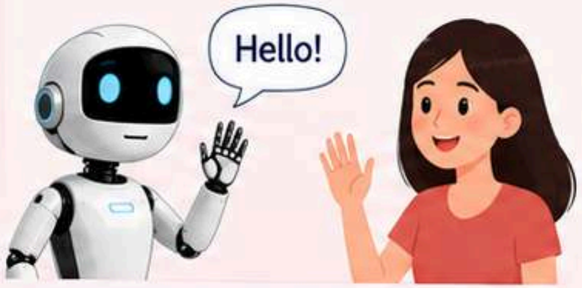


Example: Modelling human thought processes such as memory, learning and reasoning.

2 Acting Humanly

Main Question:

Can a machine behave like a human?

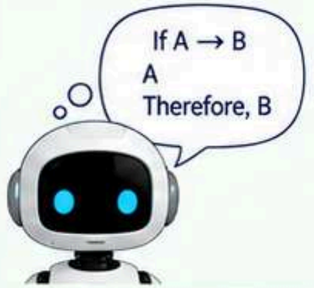


Example: Chatbots, humanoid robots, or systems that can pass the Turing Test by behaving like humans.

3 Thinking Rationally

Main Question:

Can a machine reason correctly using logical principles?



- Logical reasoning
- Mathematical proof
- Knowledge representation

Example: Using formal logic to derive correct conclusions from given information.

4 Acting Rationally

Main Question:

Can a machine choose actions that best achieve its goals?



Example: Autonomous cars, game-playing AI, or intelligent agents that select the best action based on expected outcomes.

Two Dimensions

These four views can be grouped into two dimensions:

Human-centred approaches

- Thinking humanly
- Acting humanly

Rational approaches

- Thinking rationally
- Acting rationally

Modern AI mainly focuses on **rational agents,** although ideas from all four approaches remain important.

Each view offers a **different perspective,** and together they help us understand what AI is and what it can become.

Many views. One exciting field.

Intellectual Foundations of Artificial Intelligence

Artificial intelligence did not develop from computer science alone.
It combines ideas from many disciplines, each contributing important concepts about **intelligence, learning, reasoning, decision-making** or **communication**.

> "AI is a meeting point of many fields of knowledge."

Different ideas. A smarter tomorrow.

Philosophy

Questions about the nature of mind, knowledge, reasoning and what it means to be intelligent.

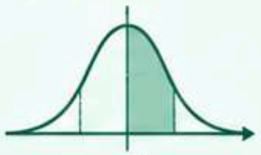
- Nature of mind
- Knowledge and belief
- Ethics and morality
- What is intelligence?



Mathematics

Provides the formal tools for representing and manipulating information.

- Logic
- Probability
- Statistics
- Optimization
- Linear algebra
- Algorithms



$$P(A) = \int_A p(x)\,dx$$

Economics

Contributes ideas about rational decision-making and the value of actions.

- Utility and value
- Decision theory
- Game theory
- Resource allocation
- Incentives



Choose the option with highest expected value.

Neuroscience

Studies the brain and nervous system, inspiring models of learning and information processing.

- Neural networks
- Brain structure
- Learning mechanisms
- Perception
- Memory



Psychology

Provides insights into human cognition, behaviour and learning processes.

- Perception
- Memory
- Problem solving
- Learning
- Human behaviour



Computer Engineering

Provides the hardware and computing systems needed to build AI.

- High-performance computing
- Specialized hardware (e.g., GPUs)
- Sensors and robotics
- Embedded systems



Control Theory

Contributes methods for designing systems that achieve desired behaviour autormatically.

- 'Feedback systems
- Stability
- Optimal control
- Robotics
- Dynamic systems



Sense → Control → Act

Linguistics

Provides knowledge about language structure, meaning and communication.

- Syntax
- Semantics
- Pragmatics
- Natural language understanding
- Language generation



The intellectual foundations of AI show that it is an inherently **interdisciplinary field**, combining insights from many areas of human knowledge.

Many disciplines. One intelligent future.

Philosophy and AI

Philosophers have studied questions about intelligence for thousands of years. Philosophy provides fundamental ideas about the **nature of mind, knowledge, reasoning and intelligence**, many of which continue to influence AI research today.

> **Big questions today. Smarter answers tomorrow.**
>
> — Philosophy inspires AI

Timeless questions. Brighter intelligence.

Key Questions

Philosophers have asked important questions such as:

- What is **knowledge**?
- How do humans **reason**?
- What is **consciousness**?
- Can reasoning be represented using **rules**?
- Can machines possess **intelligence**?
- What is the relationship between the **mind** and the **physical brain**?



Main Contributions to AI

Philosophy contributed several important ideas:

Reasoning as a systematic process
The idea that reasoning can be represented using formal rules and logic.

Concepts of knowledge
Studied the nature, sources and limits of knowledge.

Mind and consciousness
Explored the nature of the mind and whether machines can be conscious.

Ethics and values
Raised questions about moral reasoning, responsibility and the consequences of AI.

Examples of Philosophical Questions in AI

Consciousness
Can a machine be conscious, or only simulate consciousness?

Mind–body problem
What is the relationship between the mind and the physical brain?

Machine intelligence
Is machine intelligence fundamentally different from human intelligence?

Ethics
How should intelligent systems make moral decisions? Who is responsible for their actions?

Overall Impact

- Philosophy helped shape the fundamental questions of AI.
- It influenced early AI research on logic, reasoning and knowledge representation.
- It continues to be important in modern AI, especially in areas such as ethics, consciousness and the long-term impact of intelligent machines.

Key Takeaway

Philosophy contributed the idea that **reasoning may be represented as a systematic process,** and it continues to guide important questions about the nature and limits of artificial intelligence.

AI builds on ideas from **philosophy** to explore what it means to be intelligent, and to understand the **mind, knowledge** and **reasoning** more deeply.

Ancient ideas. Modern possibilities.

Mathematics and AI

Mathematics provides many of the tools required to build intelligent systems. Mathematical methods allow AI systems to **represent information, reason, learn from data and make decisions** in uncertain and complex environments.

> **Mathematics is the language that turns ideas into intelligent systems.**
>
> — Smarter algorithms. Brighter futures.

Numbers today. Smarter AI tomorrow!

Logic

Used for reasoning, rule-based systems and knowledge representation.

- Represent knowledge using formal rules.
- Basis for logical reasoning.
- Widely used in early AI and knowledge-based systems.

Example: Rule-based reasoning

IF temperature > 38°C
AND cough = true
THEN possible infection

Logic helps machines draw valid conclusions from given facts.

Probability

Used to reason under uncertainty.

- Helps handle uncertain, incomplete or noisy information.
- Provides a way to represent degrees of belief.

Example: Medical diagnosis

Probability of disease given symptoms = 0.82

$$P(\text{Disease} \mid \text{Symptoms}) = 0.82$$

Instead of a simple Yes/No answer, the system estimates the probability of a disease.

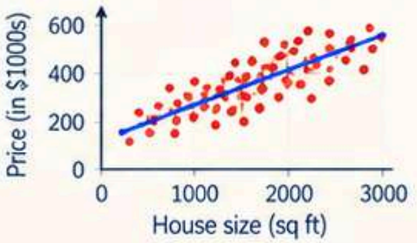
Statistics

Used to discover **patterns** from data.

- Helps analyse and interpret data.
- Used for prediction, classification and generalization.
- Essential in modern data-driven AI systems.

Example: House price prediction

Given thousands of previous house sales, a model can learn the relationship between features (size, location, etc.) and price to estimate the price of a new property.



Linear Algebra

Essential for modern machine learning and neural networks.

- Data (images, words, etc.) is often represented as vectors and matrices.
- Used in deep learning, computer vision, natural language processing and many other AI applications.

Example: Image as a matrix



$$I = \begin{bmatrix} 0.1 & 0.5 & 0.8 \\ 0.4 & 0.7 & 0.2 \\ 0.9 & 0.3 & 0.6 \end{bmatrix}$$

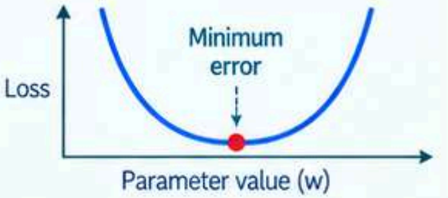
An image can be represented as a matrix of pixel values. This allows mathematical operations such as transformation, compression and feature extraction.

Optimization

Used to find the best parameter values that minimize error or maximize performance.

- Essential for training machine learning models.
- Helps find the best solution among many possible solutions.
- Common methods: gradient descent, evolutionary algorithms, linear programming.

Example: Minimizing error



Optimization algorithms adjust parameters to find the best performing model.

Key Takeaway

Mathematics provides the foundation for modern AI, enabling systems to **reason, learn from data** and **make decisions** in complex and uncertain environments.

From logic to optimization, mathematics turns ideas into **intelligent solutions.**

Same maths. Smarter machines.

Neuroscience and AI

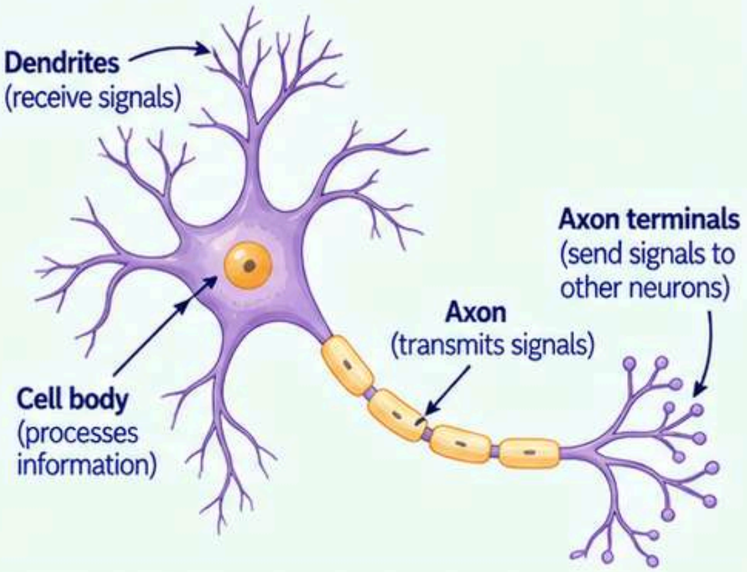
Neuroscience studies the structure and functioning of the brain and nervous system.
The human brain contains billions of interconnected neurons.
Ideas from neuroscience, especially how neurons process information, inspired **artificial neural networks**.

> **The brain inspires the algorithms that learn.**
>
> — From biology to intelligence.

Real brains. New ideas. Smarter machines.

Biological Neuron

A biological neuron receives signals from other neurons, processes them and may transmit signals to other neurons.



The human brain contains billions of neurons connected in complex networks.

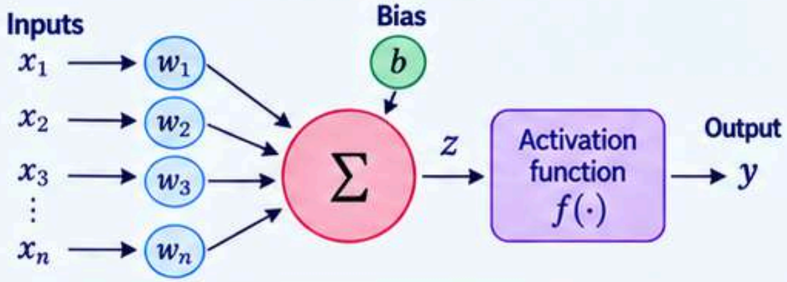
Artificial Neuron

A simplified artificial neuron receives several inputs $x_1, x_2, \cdots, x_n$. Each input has a weight $w_1, w_2, \cdots, w_n$. The inputs are combined as:

$$z = \sum_{i=1}^{n} w_i x_i + b$$

The value z is then passed through an activation function $f(\cdot)$ to produce an output:

$$y = f(z)$$



1. **Weighted sum** $z = \sum_{i=1}^{n} w_i x_i + b$
2. **Apply activation function** $y = f(z)$
3. **Produce output** y

Inspiration for AI

- The brain's structure and learning ability inspired artificial neural networks.
- Artificial neurons are simple mathematical models of biological neurons.
- Large networks of these computational units form modern neural-network architectures, such as deep learning models.



Human brain
(billions of neurons)

Important Distinction

Artificial neural networks are inspired by biological neurons, but they are highly **simplified mathematical models rather than realistic copies** of the **human brain.**

- They capture some key ideas, such as learning from examples.
- They do not replicate the full complexity of the brain, including consciousness, emotions or detailed biological processes.

Key Takeaway

Neuroscience provided important inspiration for **artificial neural networks**, helping AI systems learn from data using simplified models of brain function.

Inspired by nature. Built for the future.

Computer Engineering and AI

AI requires hardware capable of performing large amounts of computation. **Developments in computer engineering** enabled increasingly powerful AI systems. Faster processors, specialized hardware, large memory, cloud computing and high-speed networks all contribute to the development of modern AI.

> **Better hardware builds bigger possibilities for AI.**
>
> — From chips to intelligent solutions.

Powering AI today. Enabling a smarter tomorrow.

Key Advances in Computer Engineering

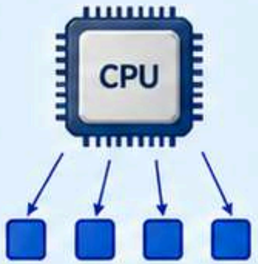
Important developments include:

- **Faster CPUs**
  More powerful general-purpose processors.
- **GPUs**
  Highly parallel processors for AI workloads.
- **Specialized AI accelerators**
  e.g., TPUs, NPUs and other dedicated hardware.
- **Larger computer memory**
  Allows training of larger models and datasets.
- **Cloud computing**
  Provides large-scale computing resources on demand.
- **Distributed computing**
  Uses multiple machines to work together.
- **High-speed networking**
  Enables fast data transfer between machines.
- **Large-scale data storage**
  Stores massive datasets for training AI models.

Why GPUs Became Important

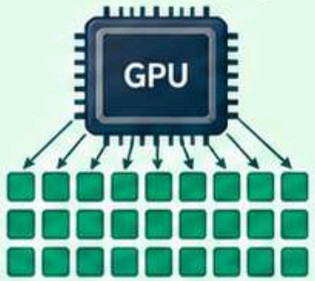
Neural-network training requires enormous numbers of matrix operations. Graphics Processing Units (GPUs) can perform many mathematical calculations in parallel, making them highly effective for deep learning.

CPU
(Sequential Processing)



Performs a few operations at a time (sequentially). Good for general-purpose tasks.

GPU
(Parallel Processing)



Performs many operations at the same time (in parallel). Ideal for matrix computations in deep learning.

As a result, GPUs became a major technology supporting modern deep learning.

Example: Matrix Operations in Training

Training a neural network involves millions or billions of matrix multiplications.

$$C = A \times B$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

These operations can be computed much faster on GPUs because they can perform many calculations in parallel.

Real-World Impact

Advances in computer engineering have enabled:

- **Larger AI models**
- Training on massive datasets
- Faster training times
- Global access via the cloud

More powerful hardware → more capable AI systems → real-world applications at scale.

Key Takeaway

Developments in computer engineering, especially GPUs and large-scale computing infrastructure, have been essential for the rapid progress of modern AI.

Stronger hardware. Smarter AI. Brighter futures.

Linguistics and Artificial Intelligence

Linguistics is the scientific study of language. It contributed knowledge about grammar, syntax, semantics, meaning, context, speech and communication. These ideas helped create the field of **Natural Language Processing (NLP)**.

> **Language is the bridge between people and ideas.**
> — From human language to intelligent systems.

Words today. A smarter tomorrow.

Key Contributions from Linguistics

Linguistics provided important knowledge about:

- **Grammar** — Rules for forming correct sentences.
- **Syntax** — How words are organized into sentences.
- **Semantics** — Meaning of words and sentences.
- **Meaning** — How meaning is constructed and interpreted.
- **Context** — How surrounding information affects meaning.
- **Speech** — How spoken language is produced and understood.
- **Communication** — How people use language to share information.

What is Natural Language Processing (NLP)?

NLP attempts to enable computers to process human language. It uses insights from linguistics, machine learning and large amounts of data to understand, interpret and generate language.

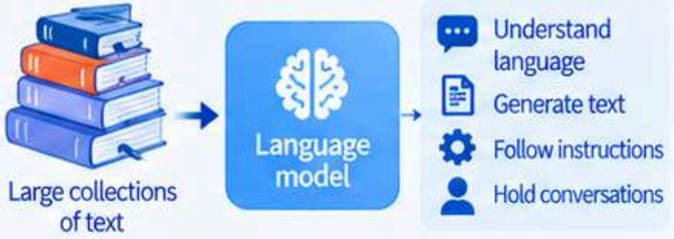
Applications of NLP

NLP is used in many real-world applications, including:

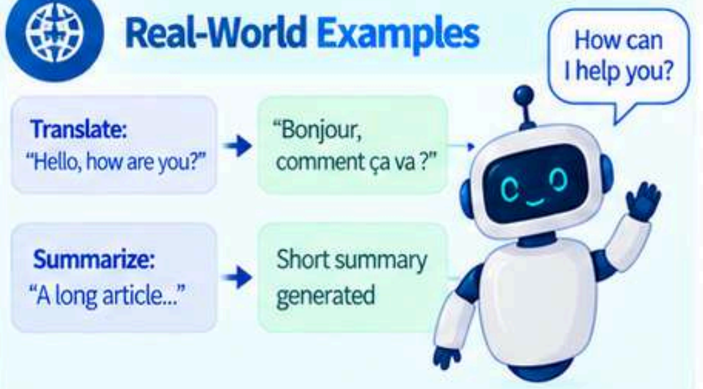
- **Machine translation** — Translate text between languages.
- **Speech recognition** — Convert spoken language to text.
- **Chatbots** — Answer questions and hold conversations.
- **Question answering** — Provide answers based on a large knowledge base.
- **Text summarization** — Create shorter versions of long documents.
- **Search engines** — Understand queries and find relevant information.
- **Generative AI assistants** — Generate human-like text, code, images and more.

Modern Language Models

Modern language models learn many linguistic patterns directly from extremely large collections of text. They use deep learning to predict, understand and generate human language with impressive ability.



Real-World Examples



Key Takeaway

Linguistics provides the foundations for NLP, helping AI systems understand, process and generate human language. This connection allows more natural and effective communication between humans and machines.

Better language. Brighter possibilities.

Dartmouth Conference and the Founding of AI

The term **Artificial Intelligence** is strongly associated with the **Dartmouth Summer Research Project on Artificial Intelligence**, held in **1956**. Researchers were highly optimistic and believed that major aspects of human intelligence could eventually be described precisely enough for machines to simulate them. This period is often considered the **formal beginning of AI as an academic field.**

> **We propose that a 2-month, 10-man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College.**
>
> — Proposal for the Dartmouth Conference

Bold ideas. Bright beginnings.

What Was the Dartmouth Conference?

The Dartmouth Summer Research Project on Artificial Intelligence was held in the summer of **1956** at Dartmouth College, USA. It brought together leading researchers to discuss the possibility of creating intelligent machines.



Dartmouth College, New Hampshire, USA (1956)

Key Researchers

Several influential researchers attended or were closely associated with the early development of AI, including:



John McCarthy
Coined the term "Artificial Intelligence"



Marvin Minsky
Pioneer in AI and cognitive science



Claude Shannon
Information theory and computation



Allen Newell
Problem-solving and cognitive modelling



Herbert Simon
Decision-making and symbolic reasoning

Optimism and Early Goals

- Researchers were highly optimistic about the potential of AI.
- Many believed that human intelligence could be described precisely enough for machines to simulate.
- They expected significant progress within a relatively short period of time.

Why It Was Important

- Introduced the term "Artificial Intelligence".
- Brought together key researchers from different fields.
- Set the research agenda for the new field.
- Marked the formal beginning of AI as an academic discipline.

Timeline

Pre-1950s
Ideas about mechanical reasoning

1950
Turing publishes "Computing Machinery and Intelligence"

1956
Dartmouth Conference (coins the term AI)

Late 1950s
Early AI programs and growing interest

A Lasting Impact

The Dartmouth Conference laid the foundation for decades of AI research and shaped the field into a major area of study.

A small conference. A big future.

Key Takeaway

The 1956 Dartmouth Conference marked the formal beginning of AI, bringing together influential researchers, coining the term "Artificial Intelligence," and setting the stage for future progress.

Ideas that shaped tomorrow.

Expert Systems: A Simple Example

Expert systems attempt to reproduce the decision-making knowledge of human specialists. They store domain knowledge as a set of rules and use an inference engine to apply those rules to the available information.

> "A simple set of rules can capture expert knowledge and provide useful advice in real-world situations."
> — AI researcher

Knowledge today. Solutions tomorrow.

Components of an Expert System

Knowledge Base
Stores facts and rules about a specific domain.

Inference Engine
Applies rules to the available information to derive conclusions.



Example Rule (Medical Domain)

A simplified rule might be:

IF
- fever = high
- cough = present
- fatigue = present

THEN
- investigate respiratory infection

How It Works

1. The system receives information (e.g., patient symptoms).
2. The inference engine checks which rules match the information.
3. If a rule matches, the system provides a conclusion or recommendation.

Areas Where Expert Systems Were Used

Expert systems were developed for many real-world applications, including:

Medicine
e.g., diagnosis and treatment recommendations

Engineering
e.g., design advice and fault finding

Finance
e.g., investment advice and risk assessment

Manufacturing
e.g., process control and production planning

Equipment Diagnosis
e.g., finding faults in complex machines and devices

Impact and Limitations

Positive Impact
- Demonstrated practical, real-world applications.
- Showed that AI could be commercially valuable.
- Attracted increased research funding.
- Renewed optimism in the field.

Limitations
- Worked mainly in narrow domains.
- Knowledge had to be encoded manually.
- Difficult to handle uncertainty and common-sense reasoning.
- Performance declined outside their specific domain.

Key Takeaway

Expert systems in the late 1970s and 1980s showed that AI could be practically useful in real-world applications, using rule-based knowledge and inference. Although limited, they helped restore interest and funding in AI.

Real knowledge. Real impact.

Why Machine Learning Became Successful

Several key developments supported the rise of machine learning. The growth of digital data, greater computing power, better algorithms and improved storage created the conditions for modern data-driven AI. Together, these advances made it possible to train more powerful models on large-scale real-world data.

> **Machine learning succeeded because the world finally had enough data, computing power and better algorithms to turn ideas into real-world impact.**
> — AI researcher

More data.
More power.
Better ideas.
Bigger impact.

1 More Digital Data

The growth of the internet, smartphones, online services, digital sensors and social networks produced enormous amounts of data.

- **The internet**
  Web pages, search queries, images, videos
- **Smartphones**
  Apps, location data, photos
- **Online services**
  E-commerce, streaming, cloud platforms
- **Digital sensors**
  IoT devices, cameras, medical sensors, vehicles
- **Social networks**
  Posts, likes, connections, multimedia

These sources generated huge and diverse datasets, providing the raw material for modern machine learning.

2 Greater Computing Power

Processors became much faster and cheaper, allowing complex models to be trained on large datasets.



- More powerful CPUs and GPUs
- Massively parallel processing
- Cloud computing made high-performance computing accessible

Increased computing power dramatically reduced the time and cost needed to train machine learning models.

3 Better Algorithms

Researchers developed improved techniques that could learn patterns from data more effectively.

- **Classification**
  Predicting categories (e.g. spam or not spam)
- **Regression**
  Predicting numeric values (e.g. house prices)
- **Clustering**
  Grouping similar data (e.g. customer segments)
- **Probabilistic modelling**
  Handling uncertainty (e.g. Bayesian methods)
- **Neural networks**
  Learning complex patterns (e.g. deep learning)

These algorithmic advances allowed machines to learn useful patterns from large amounts of real-world data.

4 Improved Storage

Large datasets became practical to store and process thanks to advances in storage technologies.



- Larger and cheaper storage devices
- Distributed storage systems (e.g. cloud)
- Efficient data processing frameworks (e.g. Hadoop, Spark)
- Ability to store and process structured and unstructured data (images, text, video, etc.)

Improved storage made it feasible to keep and work with massive datasets, which are essential for modern AI.

Key Takeaway

The combination of more digital data, greater computing power, better algorithms and improved storage created the conditions for modern data-driven AI and the success of machine learning.

Real progress happens when many advances come together.

The Deep-Learning Era

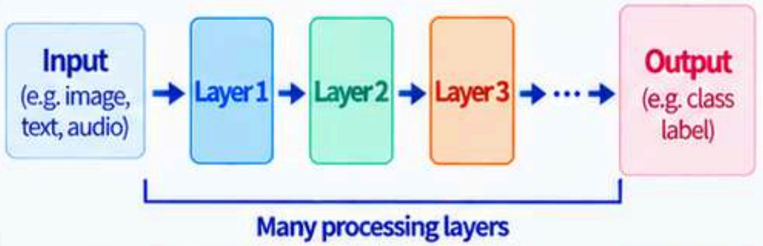
Deep learning is a branch of machine learning that uses neural networks with many processing layers. It became especially influential during the 2010s, leading to major advances in areas such as image recognition, natural language processing, speech recognition and more.

> **Deep learning succeeded because it can automatically learn useful representations from large amounts of data, allowing remarkable progress on real-world problems.**
>
> — AI researcher

Bigger data. Deeper networks. Bigger possibilities.

1 What Is Deep Learning?

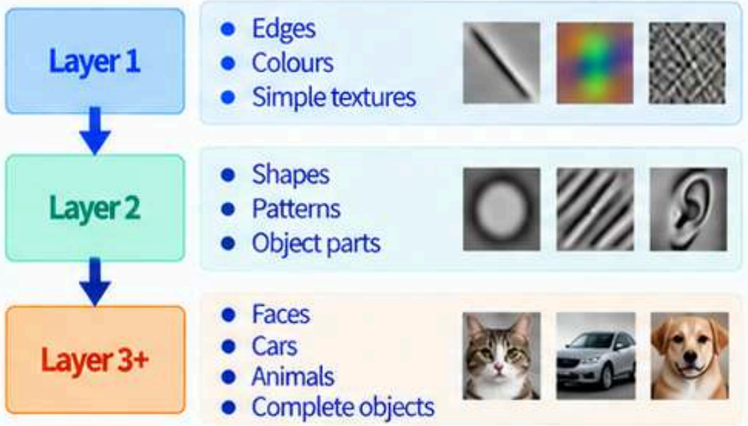
Deep learning uses neural networks containing many processing layers (hence "deep").
Each layer transforms the input data into a more useful representation.



Different layers can learn increasingly complex representations of the input data.

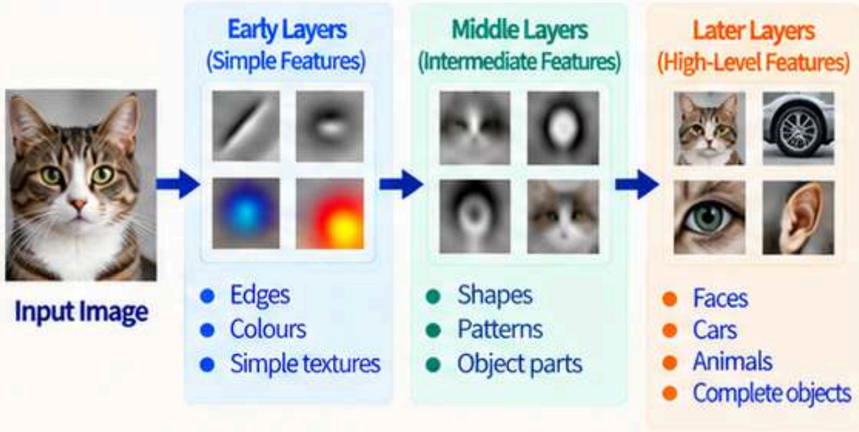
2 How Layers Learn Representations

Each layer learns to extract useful features from the output of the previous layer. Early layers learn simple features, and later layers learn more complex features.



3 Example: Image Recognition

For an image, deep learning can automatically learn a hierarchy of features from simple to complex.



4 Why Deep Learning Became Successful

Large Datasets
Massive amounts of digital data (e.g. images, text, video).

Powerful Computing
Faster and cheaper GPUs and other processors.

Improved Algorithms
More effective neural network architectures and training techniques.

Better Tools
Frameworks like TensorFlow and PyTorch made deep learning easier to use and scale.

5 Impact of the Deep-Learning Era

- Major improvements in image and speech recognition.
- Breakthroughs in natural language processing (e.g. large language models).
- New applications in healthcare, finance, robotics and more.
- Brought AI into real-world products and everyday life.
- Significantly increased interest, investment and research in AI.

Key Takeaway

Deep learning can automatically learn useful representations from large amounts of data using many processing layers. This ability to discover complex patterns without manual feature engineering was a key reason for its success in the 2010s and beyond.

From data to discovery. From layers to real-world impact.

What Is Generative AI?

Generative AI is a type of AI that can create new content, such as text, images, audio, video and code, rather than only making predictions or classifications. This has changed AI from something often used in the background into a technology that people can interact with directly.

> **Generative AI is a major shift because it lets people create, explore and bring ideas to life using natural language, images and other forms of input.**
> — AI researcher

From predictions to creation. A more interactive AI.

1 Traditional Predictive AI

Traditional AI systems often answer specific questions by analysing data and making predictions or classifications.

Fraud detection
"Is this transaction fraudulent?"

Demand forecasting
"What will tomorrow's demand be?"

Recommendation systems
"Which product should be recommended?"

These systems predict an answer from existing data rather than creating new content.

2 Generative AI Creates New Content

Generative AI can produce original content in different forms, including:

- **Text** — Articles, emails, stories, summaries
- **Images** — Photos, illustrations, artwork
- **Audio** — Voice, sound effects, spoken responses
- **Music** — Songs, melodies, background music
- **Video** — Short clips, animations, full videos
- **Computer code** — Programs, scripts, websites
- **Designs** — Logos, posters, presentations, UI mockups

3 Example: Generating an Email

Given the instruction:
"Write a professional email requesting a meeting next Monday."
a generative model can produce a complete email:

Generated Email (Example)

Subject: Request for Meeting Next Monday

Dear [Name],

I hope you are doing well. I would like to request a meeting next Monday to discuss [topic or purpose]. Please let me know a convenient time for you.

Kind regards,
[Your Name]



Instead of simply classifying the instruction, generative AI understands the request and creates a new piece of content.

4 Why It Matters

More interactive
People can communicate with AI using natural language.

Wider creativity
Helps users create, explore and bring ideas to life.

Real-world impact
Used in education, business, healthcare, entertainment and more.

Key Takeaway

Generative AI creates new content across many forms, making AI more accessible and interactive, and opening up a wide range of real-world applications.

New ideas. New content. A bigger future.

AI Agents and Tool Use

Modern AI research increasingly focuses on AI agents. An agent can perform a sequence of steps to achieve a goal, using reasoning, planning and tool use. This moves AI from simply answering questions toward performing real-world, multi-step tasks.

> "AI agents can perceive, plan, use tools, take actions and adjust based on results, allowing them to complete complex tasks in the real world."
> — AI researcher

More than answers. Real actions. Real results.

1 What Is an AI Agent?

An AI agent is a system that can:

- Understand a goal
- Plan a sequence of steps
- Take actions (often using external tools)
- Observe the results and adjust if needed



An agent does more than produce a single prediction. It can autonomously work towards a goal through multiple steps.

2 The Agent Loop

A typical agent follows a cycle:



3 Tools Agents Can Use

AI agents can interact with many external systems, including:

- **Search systems** (e.g. web search)
- **Databases** (e.g. company or research data)
- **Software tools** (e.g. calculators, code editors)
- **APIs** (e.g. weather, maps, booking services)
- **Computer applications** (e.g. email, calendars)
- **Robots** (e.g. physical actions in the real world)

4 Example: Planning a Three-Day Business Trip

Goal: "Plan a three-day business trip."

1. **Determine travel requirements** (e.g. destination, dates, budget).
2. **Search transportation options** (e.g. flights, trains).
3. **Compare hotels** (e.g. location, price, reviews).
4. **Organize a schedule** (e.g. meetings, travel times).
5. **Produce an itinerary** (e.g. complete plan with all details).

5 Real-World Examples

Travel planning
Book flights, hotels and create itineraries.

Research assistant
Search and summarise information from multiple sources.

Software development
Use coding tools, run tests, debug and deploy.

Data analysis
Access datasets, run analyses and generate reports.

Online shopping
Find products, compare options and make purchases.

Task automation
Use applications to complete repetitive tasks.

6 Benefits and Challenges

Benefits

- Can complete complex, multi-step tasks
- More useful in real-world situations
- Can integrate multiple tools and data sources
- Increases productivity
- Expands what AI can do

Challenges

- May still make mistakes
- Requires careful oversight
- Security and privacy risks
- Dependence on external tools
- Needs clear goals and constraints

Key Takeaway

AI agents can plan, use tools, take actions and adjust based on results, enabling them to perform complex, real-world tasks rather than just answering questions.

From answers to actions. A more capable AI.

CSC262
Artificial Intelligence
BS (Computer Science)

Fall 2026

"Building Intelligent Solutions
for a Better Tomorrow"



COMSATS University Islamabad
Faculty of Information Science and Technology

Learn | Think | Build | Innovate

Intelligent Agents

Intelligent Agents: PEAS & Rationality

What is an Agent?

An **agent** is an entity that:

- **Perceives** its environment through **sensors**
- **Acts** upon the environment through **actuators**

Agent–Environment Interaction



Environment → Sensors → Agent → Actuators → Environment

Examples



Human



Robot



Self-driving car



Software assistant



Vacuum-cleaning robot

AIMA introduces agents in terms of their perception of and actions in an environment.



Percepts and Percept Sequence

A **percept** is the information received by an agent at a particular moment.

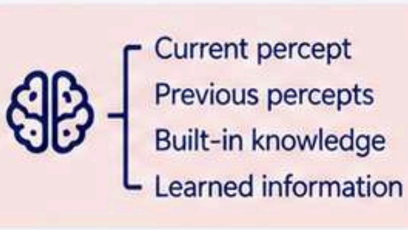
Example for a vacuum agent:

```
Location = Room A
Dirt detected = Yes
```



An agent may choose its next action using:

- Current percept
- Previous percepts
- Built-in knowledge
- Learned information



Percept Sequence (Vacuum Agent)

Time t_1	Time t_2	Time t_3	Time t_4
Location = A Dirt = **Yes**	Location = A Dirt = **No**	Location = B Dirt = **Yes**	Location = B Dirt = **No**

Percept sequence
(complete history of everything the agent has perceived)

A **percept sequence** is the complete history of everything the agent has perceived.

AIMA describes an agent's behavior in terms of how percept sequences are mapped to actions.



AI Model vs. AI Agent

These two terms should not be confused.

AI Model

A model mainly performs computation such as:

Input
(e.g., a question)
→ **Model** →
Output
(e.g., an answer)

Example:

Student:
"Explain binary search."
→
Model:
generates an explanation.

AI Agent

An agent can perceive, reason, take actions, and use tools to achieve goals in an environment.

Perceive
(get information)
→ **Reason**
(plan, decide)
→ **Act**
(use tools, take actions)
→ **Observe results**
(get feedback)

Continue towards goal

Example: AI Coding Agent

Code files + terminal output
→ **AI agent** →
Edit files + run commands + run tests

An agent is more than simply generating text. It can **observe** → **reason** → **use tools** → **observe results** → **continue**.

Intelligent Agents

Intelligent Agents: PEAS & Rationality



The Modern Agent Loop

A useful way to understand today's AI agents is:

1 Goal

"Fix the login bug."

Set the objective for the agent.

2 Perceive

Agent reads:

- Source files
- Error messages
- Test results

Gather information from the environment.

3 Decide

"I should inspect the authentication function."

Reason about the next action.

4 Act

Agent:

- Opens file
- Changes code
- Runs test

Use tools to take action in the environment.

5 Observe

"Test failed."

Get feedback from the environment.

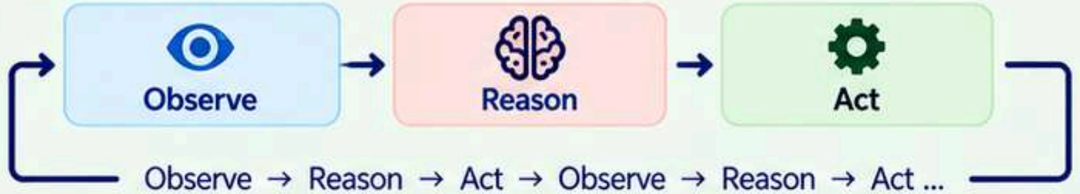
6 Decide Again

The process repeats.

Use the new information to decide the next step.

Repeat until the goal is achieved

Agent Loop





Key Point

Modern agent systems are specifically designed to support these repeated **multi-step interactions** with tools and environments.

An agent is more than simply generating text. It can **observe** → **reason** → **use tools** → **observe results** → **continue**.



Actuators in Modern AI Agents

An **actuator** allows an agent to change its environment.

Physical Agent

A robot may use:


Wheels


Motors


Robotic arm



Software AI Agent

Its actuators may be:


Write a file


Edit code


Run a terminal command


Search the web


Call an API


Send a message

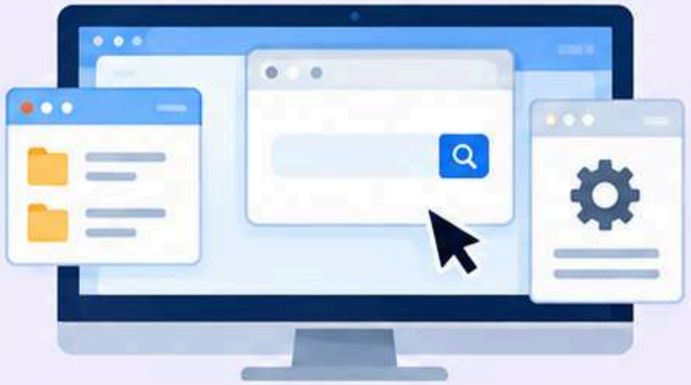

Fill a web form


Click a button



Computer-Using Agents

Agents can interact with graphical software interfaces rather than only producing text responses.



They can open applications, click buttons, fill forms, and operate software just like a human user.



In all cases, **actuators** allow the agent to change its environment.

Intelligent Agents

Intelligent Agents: PEAS & Rationality

PERCEIVE
REASON
ACT
LEARN

PEAS Example: AI Research Agent

Task: **"Compare solar and wind power using recent reliable information."**



P Performance
What does "good" mean?

- Correct information
- Relevant sources
- Recent information
- Clear explanation
- Proper citations

E Environment
Where will it work?

- Internet
- Search engines
- Websites
- Research papers
- Documents

A Actuators
What is it allowed to do?

- Search
- Open pages
- Read documents
- Analyze information
- Create report

S Sensors
What information can it access?

- Search results
- Webpage text
- Tables
- Documents
- User instructions

Search (relevant sources) → **Read** (web pages / papers) → **Analyze** (compare information) → **Create Report** (with citations)

Modern agent platforms commonly expose tools such as **web search**, **file access**, **computer interaction**, **and code execution** to the model.

Intelligent Agents

Intelligent Agents: PEAS & Rationality

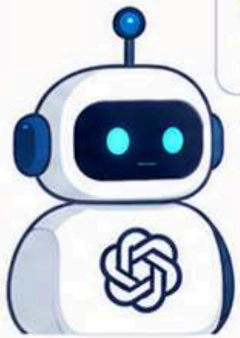


Chatbot vs. Research Agent

Consider the question: *"What is Pakistan's current inflation rate?"*

Simple Chatbot Approach

Use information stored or learned previously.



What is Pakistan's current inflation rate?

Pakistan's inflation rate is around 9.2%.
(Based on my knowledge)

Problem:
The information may be outdated.

A simple chatbot may give a confident answer, but it might not be current.

Agent Approach

Uses tools to find and verify current information.

1. Understand that current data is required
2. Search the web
3. Find a reliable current source
4. Read the data
5. Compare sources if necessary
6. Give an answer with evidence



Let me check the latest data from reliable sources.

Search → Read & Compare →

Latest Data

Pakistan's inflation rate is 11.1% (July 2024).

Source: State Bank of Pakistan

Provides a current answer with evidence.

Key Lesson | Sometimes **using a tool** is itself the **rational action.** | Agents can decide when to use tools, search for information, and take multiple steps to achieve a goal.



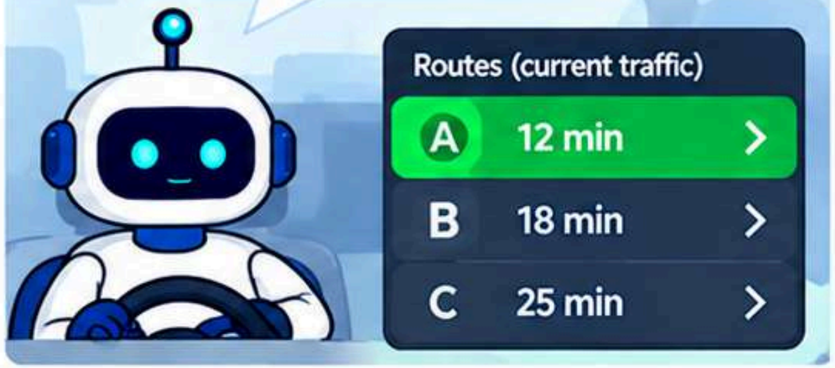
Rationality vs. Omniscience

Rational agents make the best decision based on what they know, not on perfect knowledge of the future.

Rational Agent

Uses available information.

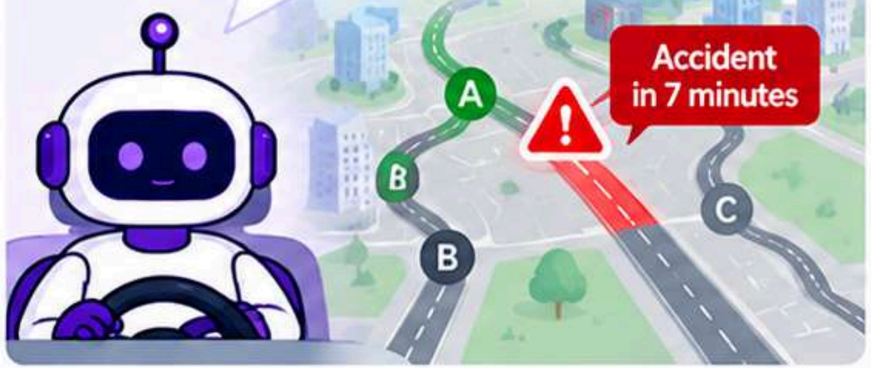
"Traffic data shows Route A is fastest."



Omniscient Agent

Would somehow know everything that will happen.

"There will be an accident on Route A in seven minutes."



Key Points

- Real agents are not omniscient.
- They operate under uncertainty.
- They make the best decision using the information available at that moment.

Therefore:
Rationality ≠ Knowing the future

Key Takeaway

A rational agent chooses the best action with limited information.
This is different from an omniscient agent, which would know everything that will happen.



AI Can Write Code. Can You Verify It?

AI can generate large amounts of code very quickly. But verification, evaluation, and judgment are still human responsibilities.

AI Generates Code

Suppose an AI agent generates 500 lines of code in 20 seconds.

AI-generated code

```
# ... 500 lines of code ...
def process_data():
    data = load_data()
    result = []
    for item in data:
      if validate(item):
        result.append(transform(item))
    return result
```

500 lines of code in 20 seconds. Amazing.

But Now Someone Has to Answer...

- Is it correct?
- Is it secure?
- Is it maintainable?
- Does it satisfy the requirement?
- Does it destroy existing functionality?
- What happens at scale?
- Can we trust the dependencies?
- Are there hidden edge cases?

Why Verification Matters

AI assistance is widespread, but developers' continuing skepticism about output accuracy shows why verification remains an engineering skill.



Verification is not optional.
It is an essential engineering skill.

New Valuable Skill

Not just generation — but evaluation, verification, and improvement.

- Read and understand AI-generated code
- Verify correctness
- Check security and reliability
- Evaluate design and maintainability
- Test and find edge cases
- Identify potential issues
- Improve and refactor when needed

Not just generation. Verification creates value.

Key Takeaway

AI can write code. But you need the knowledge and judgment to verify it, improve it, and ensure it works correctly in the real world.

LEARN
REASON
BUILD A
BRIGHTER
TOMORROW

The Big Picture

ARTIFICIAL INTELLIGENCE
PEOPLE IDEAS INTELLIGENT SYSTEMS

Two questions guide agent design

Question 1

What kind of world does the agent operate in?

→ Nature of the environment



Question 2

What information should the agent use to choose actions?

→ Structure of the agent

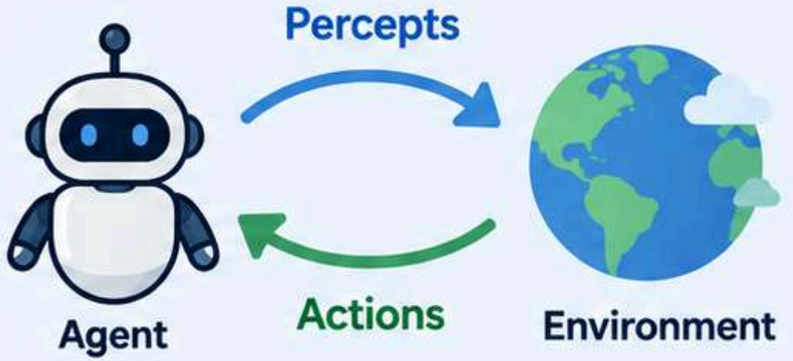


Environment → Agent Design → Behavior

Properties of Task Environments

A task environment can be classified along several dimensions:

Property	Possibilities
Observability	Fully / Partially
Predictability	Deterministic / Stochastic / Strategic
Dependence over time	Episodic / Sequential
Change during reasoning	Static / Dynamic / Semidynamic
State/action space	Discrete / Continuous
Number of agents	Single / Multiagent
Environment model	Known / Unknown



These properties help determine **how difficult the task is and what kind of agent architecture is appropriate.**

The environment often determines the kind of intelligence the agent needs.

Fully Observable vs. Partially Observable

How much of the environment can the agent see?

Fully Observable

The agent's sensors provide the information relevant to choosing an action.

Examples

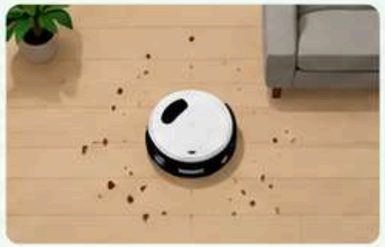
- Chess board
- Crossword puzzle
- Simple vacuum world with complete sensors



Chess



Crossword puzzle



Vacuum world (complete sensors)

Partially Observable

Important information is hidden, missing, uncertain, or noisy.

Examples

- Self-driving vehicle
- Robot using cameras
- Poker
- AI research agent browsing incomplete web information



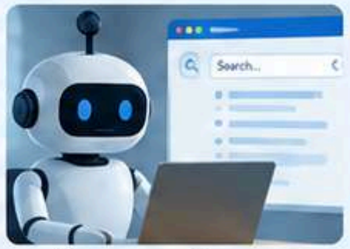
Self-driving vehicle



Robot using cameras



Poker



AI research agent (incomplete information)

A partially observable agent may need **memory or an internal model.**

Single-Agent vs. Multiagent

Does the agent have to consider other agents in the environment?

In multiagent environments, other agents also choose actions, which can affect the outcome.

Single-Agent Environment

Only one relevant decision-making agent.

Examples

- Crossword solving
- Image classification
- Solving a mathematical puzzle



Crossword solving

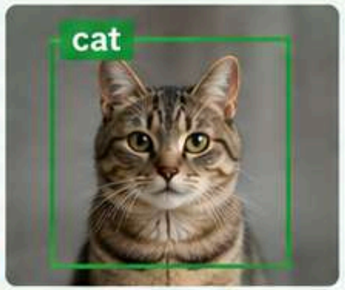
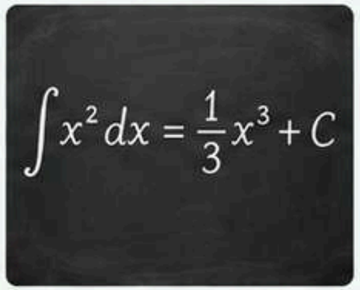


Image classification



Mathematical puzzle

Multiagent Environment

Other agents also choose actions.

Examples

- Chess
- Traffic
- Online auctions
- Robot teams
- Multiple AI agents collaborating on software



Chess



Traffic



Online auctions



Robot teams



AI agents collaborating

Multiagent environments may be:

- Cooperative
- Competitive
- Mixed

Chess is treated as competitive, while traffic includes both cooperative and competitive aspects.

In multiagent environments, the agent must consider that other agents are also making decisions.

Modern Multi-Agent AI

Multiple AI agents can work together to build a large application.

Imagine developing a large application using several AI agents:

Agent 1
Implements backend

Agent 2
Builds frontend

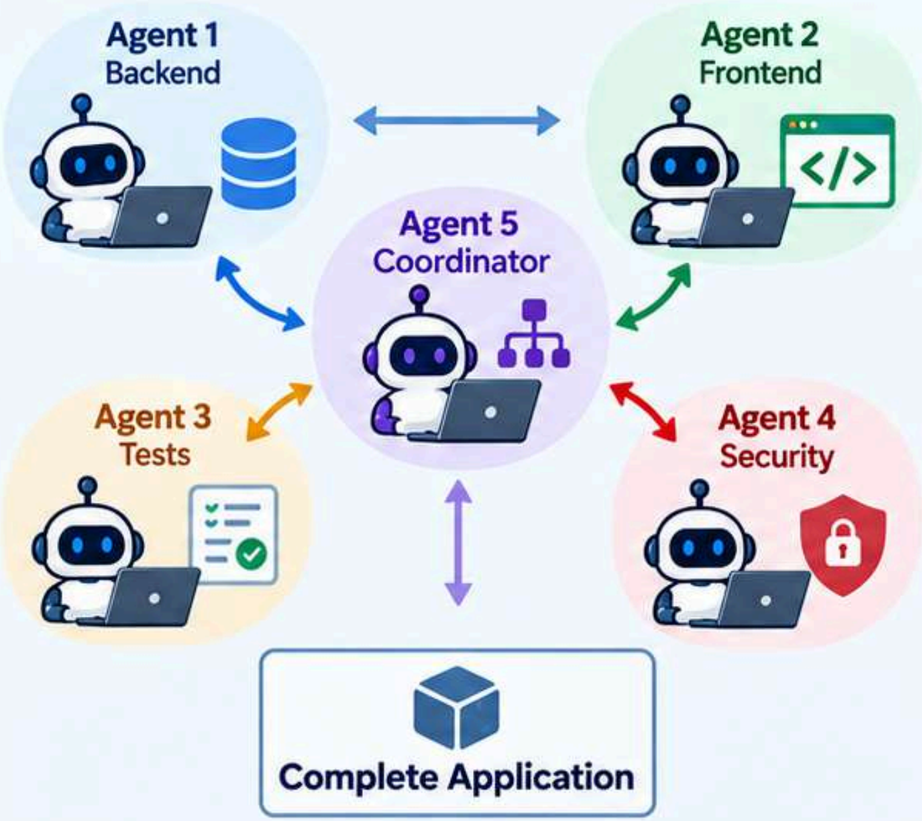
Agent 3
Writes tests

Agent 4
Reviews security

Agent 5
Coordinates the work

Working Together

Agents collaborate to build the complete application.



Agents may need to:

Communicate
Share information and updates

Divide tasks
Assign and manage work

Share context
Keep a consistent understanding

Detect conflicts
Identify and resolve issues

Coordinate changes
Ensure changes work well together

Modern coding systems increasingly support parallel or coordinated agents working on different tasks.

As AI agents work together, the environment becomes more complex, and coordination is essential.

Deterministic vs. Stochastic Environments

Does the same action always lead to the same result?

Same action, different outcome?

Deterministic Environment

Given the current state and action, the next state is completely determined.

Result = f(current state, action)

Example: Simple vacuum world

If the agent is in a given location and chooses "suck", the next state is always the same (dirt is removed).



State: Dirty

Next state: Clean

Useful when outcomes are predictable and consistent.

Stochastic Environment

The same action may produce different outcomes.

Result = probability distribution over possible next states

Examples



Driving

Same steering action may lead to different outcomes (e.g., other cars, road conditions).



Robot manipulation

Grasping may succeed or fail due to noise, slippage, or uncertainty.



Medical treatment

Same treatment can have different effects on different patients.



Financial markets

Same action (e.g., buy) may lead to different outcomes due to market uncertainty.

Strategic Environment

Uncertainty mainly results from another agent's choices.

Outcomes depend on what other agents do (in addition to the current state and action).

Examples



Chess

Outcome depends on opponent's moves.



Traffic

Other drivers also choose actions.

AIMA uses strategic environments when uncertainty is primarily due to other agents.

Deterministic, stochastic, and strategic environments require different agent designs and reasoning methods.

Episodic vs. Sequential Environments

Do current actions affect future situations?

Some tasks are one-shot, others require long-term thinking.

Episodic Environment

Each decision is mostly independent of previous decisions.

Percept → Action → Episode ends

Examples



Classifying one image



Detecting whether one email is spam



Inspecting individual manufactured components

Sequential Environment

Current actions influence future situations.

Examples



Chess
Each move affects the future game.



Driving
Current actions influence future traffic situations.



Planning
Early decisions constrain later options.

```
def compute(x):
    return x * 2

# modify this function
# affects later behavior
```

Software modification
Changes today affect future behavior.



Robot navigation
Actions influence future locations and observations.

Key Point from AIMA

Sequential problems require considering longer-term consequences, not just the immediate next action.

Why It Matters

- Affects how the agent is designed
- Influences the need for planning and search
- Impacts learning from experience
- Sequential environments are often more challenging

Understanding whether a problem is episodic or sequential helps us choose the right agent architecture and reasoning method.
Many real-world AI applications, such as autonomous driving and software development, are sequential.

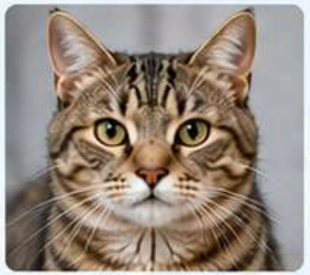
Classification AI vs. Agentic AI

Same AI technology, different types of environments.

Some tasks are one-shot, others require a sequence of decisions.

Image Classifier

Classify a single image.



Input Image

→ Image Classifier (model) →

Output

Cat	0.92
Dog	0.06
Other	0.02

Prediction: Cat

Environment Type

Mostly episodic

- Each image is classified independently.
- The next image does not depend on previous ones.
- The agent does not need to plan ahead.

Examples

- Image classification
- Spam detection (one email)
- Quality inspection (one item)

Coding Agent

Fix a bug or implement a feature using multiple steps.

1. **Read files** — Understand the codebase
2. **Modify code** — Make changes
3. **Run tests** — Check if it works
4. **Observe failure** — See error messages and logs
5. **Modify again** — Improve and repeat

Repeat until the task is solved

Environment Type

Sequential

- Current actions influence future situations.
- The agent must consider longer-term consequences.
- One decision changes what the agent encounters later.
- Often requires planning, exploration, and iteration.

Examples

- Fixing bugs in a codebase
- Implementing a new feature
- Software refactoring
- Running and interpreting tests
- Using tools (terminal, debugger, etc.)

Classification AI is often episodic, while agentic AI for real-world tasks is sequential.

Sequential problems require reasoning about longer-term consequences, not just the next action.

Static vs. Dynamic Environments

Does the environment change while the agent is deciding?

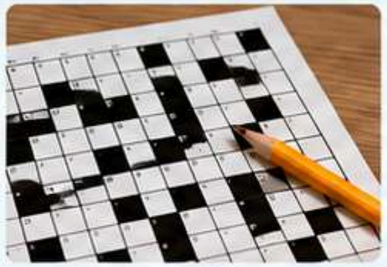
In some environments, time keeps running. The world may change while the agent thinks.

Static Environment

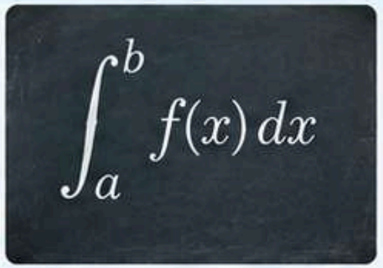
The environment does not change while the agent is deciding.

The agent can take time to compute the best action without worrying about the world changing.

Examples



Crossword puzzle
The puzzle does not change while you think.



Offline mathematical problem
The problem remains the same while you reason.

Good for deep computation and search.

Dynamic Environment

The environment may change while the agent is reasoning.

The agent must make decisions in real time, often with incomplete and changing information.

Examples



Driving
Other vehicles and road conditions keep changing.



Stock trading
Market conditions change continuously.



Robot navigation
The robot's surroundings may change.



Cybersecurity monitoring
New threats can appear at any time.

Semidynamic Environment

The environment itself may remain unchanged, but the performance measure changes with time.

Actions do not change the state of the environment, but time affects the outcome (e.g., a penalty for taking too long).

Example



Chess with a clock
The board does not change by itself, but time keeps running, and taking too long reduces performance.

Understanding whether an environment is static, dynamic, or semidynamic helps us choose the right agent design.
Real-world AI systems, such as autonomous vehicles and online services, often operate in dynamic or semidynamic environments.

Discrete vs. Continuous Environments

Are the states and actions separate or do they vary smoothly?

Many real-world problems involve continuous values.

Discrete Environment

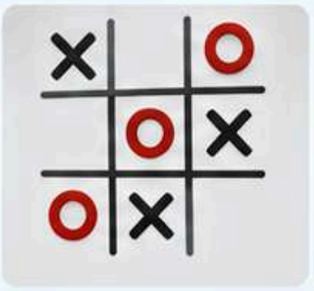
Finite or clearly separated states and actions.

The environment has a countable set of states and actions. Changes happen in distinct steps.

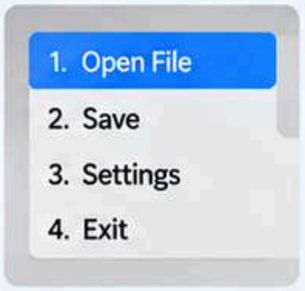
Examples



Chess moves
A finite set of legal moves in each state.



Tic-tac-toe
Discrete board states and actions.



Menu selections
Choose from a fixed set of options.

Continuous Environment

States and actions vary over a range of values.

The environment is described by continuous variables that can take any value within a range.

Examples



Vehicle speed
Can take any real value (e.g., 0–200 km/h).



Robot joint angle
Can be any angle (e.g., 0–360°).



Distance
Any real-valued distance.



Temperature
Varies continuously over a range.

Key Point

Real physical environments, such as driving or robotics, are often largely continuous, even if we sometimes approximate them as discrete in practice.

Implications for AI

- Continuous environments require sensing, estimation, and control.
- Actions must be precise (e.g., motor commands).
- Often combined with uncertainty and real-time constraints.

Understanding whether a problem is discrete or continuous helps us choose the right representation, algorithms, and control methods.

Many real-world AI systems (e.g., autonomous vehicles, robots) operate in continuous environments.

Known vs. Unknown Environments

Does the agent know how its actions affect the environment?

Knowing the current situation is not the same as knowing how the world works.

Known Environment

The agent knows how its actions affect the environment.

The rules (state transitions) are given or can be fully derived. The agent can predict the outcome of its actions.

Examples



Chess

The rules are known. The agent knows the result of each move.



Simple vacuum world

The agent knows what happens when it moves, cleans, etc.



Mathematical problem

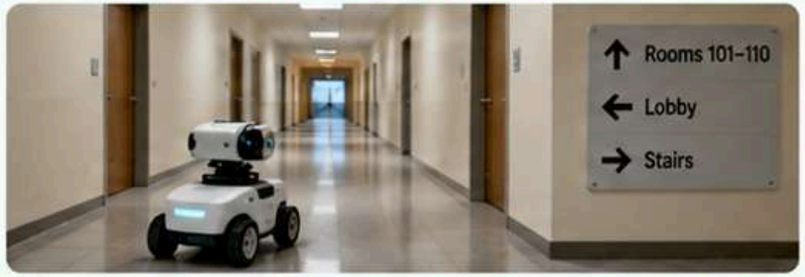
The effects of actions (e.g., computing a step) are known.

Unknown Environment

The agent must discover how the environment behaves.

The agent does not initially know the rules or the effects of its actions. It must learn by interacting with the environment.

Example: Robot in a New Building



A robot enters a new building. It may observe a corridor clearly but still not know:

- Where it leads
- Which doors are locked
- What lies around the corner

The agent must explore and learn the environment through experience.

Key Distinction

Observable = What is happening?

Can the agent see or sense the current state of the environment?

Known = Do I understand how this world works?

Does the agent know how its actions affect the environment (state transitions)?

An environment can be observable but unknown.

Example: A robot can see the corridor (observable) but still not know where it leads (unknown).

Why It Matters

- Unknown environments require exploration and learning.
- The agent must build a model of how the world works.
- Common in real-world AI systems (robotics, games, new domains).

Observability tells you what is happening. Knowledge tells you how the world works.

Many real-world AI agents must both perceive the current state and learn the effects of their actions.

Case Study: AI Coding Agent

Analyzing an agent that finds and fixes a real bug in a codebase.

More than code generation — modern agents can explore, test, and iterate.

Task

"Find and fix the authentication bug."

Example Context

A user reports they cannot log in. The agent is given a repository, issue description, and test environment.

```
Issue #482
Login fails with 401 error

Steps to reproduce:
1. Run the app locally
2. Try to log in with a valid account
3. Observe 401 Unauthorized
```

The agent must understand the code, find the root cause, make changes, and verify the fix.

How the Coding Agent Works

The agent uses tools and feedback to solve the task iteratively.

1. **Inspect repository**
   Search and read relevant files, error logs, and documentation.

```
src/
  auth.py
  routes.py
  test_auth.py
  ...
```

2. **Modify code**
   Edit files to fix the bug.

```
- return None
+ return user
```

3. **Run tests**
   Execute unit and integration tests.

```
$ pytest
✓ 42 passed
  0 failed
```

4. **Observe results**
   Analyze failures or confirmations.

All tests passed!

5. **Iterate if needed**
   Make further changes until the issue is resolved.

✓ Fix committed

Possible Classification

How this task maps to environment properties:

Partially observable	Requirements and system details may be missing.
Sequential	Each code change affects later tests and outcomes.
Partly stochastic	Dependencies, external services, or test environments may vary.
Mostly discrete actions	Edit, test, search, commit, etc.
Dynamic	The repository or external services may change while the agent is working.
Potentially multiagent	Humans or other agents can modify the code, create issues, or review changes.

Key Insight

Modern coding agents can inspect repositories, modify files, execute commands and tests, and iterate on results rather than merely generating a code snippet.

Coding agents operate in complex, real-world environments that combine multiple properties.

Understanding these properties helps us design better agents and set realistic expectations.

Case Study: AI Research Agent

Producing an evidence-based report on a current technology.

Find information.
Analyze evidence.
Synthesize insights.
Produce a reliable report.

Goal

"Produce an evidence-based report on a current technology."



The web contains incomplete, conflicting, and changing information. The agent must evaluate sources and reason about uncertainty.

Environment: Information Sources

The agent interacts with a real-world, open, and constantly changing information environment.

Search results
e.g., Google, Bing, academic search

Web pages
e.g., news sites, blogs, documentation

PDFs
e.g., research papers, white papers, reports

Databases
e.g., academic databases, industry data, statistics

Agent Actions

The agent uses tools to gather, process, and synthesize information.

1. **Search** — Find relevant sources.
2. **Open sources** — Read and understand content.
3. **Extract data** — Pull key facts and figures.
4. **Run analysis** — Compare, summarize, and analyze.
5. **Write report** — Synthesize findings with proper citations.

Environment Properties

This task has multiple important properties.

Partially observable	Not all relevant information is visible or accessible.
Sequential	Each action (e.g., reading, analysis) affects later steps.
Dynamic	Web content changes over time. New information can appear at any time.
Uncertain (stochastic)	Sources may be incomplete, conflicting, or unreliable. Search results can vary.

Why These Properties?

- The web contains incomplete information.
- Different sources may conflict.
- Information can change quickly.
- The agent must decide which sources to trust.
- Final output depends on a sequence of steps (search → read → analyze → write).

AI research agents operate in complex, uncertain, and changing environments.
They must find, evaluate, and synthesize information from multiple sources to produce reliable, evidence-based reports.

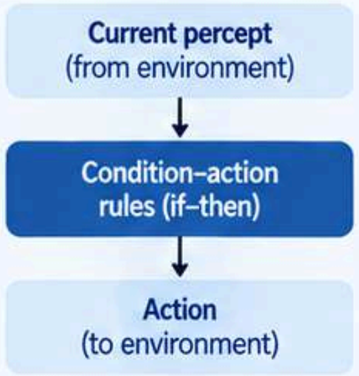
The Main Agent Types

AIMA introduces a progression of agent designs, from simple reflex behavior to learning agents.

Different problems require different agent designs. More capability usually comes with more complexity.

1 Simple Reflex Agent

Responds directly to the current percept using condition–action rules.



Examples

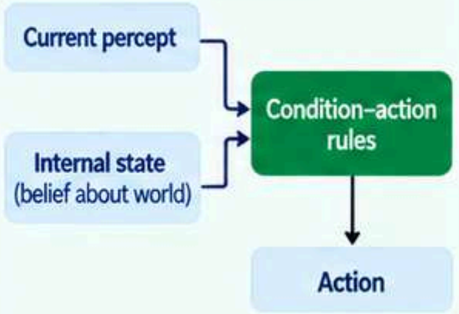
- Thermostat
- Simple vacuum cleaner

Key idea

Act based only on the current percept, ignoring history.

2 Model-Based Reflex Agent

Maintains an internal state to handle partial observability.



Examples

- Robot vacuum (keeps a map)
- Autonomous drone (tracks state)
- Game agent (remembers unseen parts of the world)

Key idea

Use an internal state to represent the world, not just the current percept.

3 Goal-Based Agent

Uses goals and considers future consequences of actions.



Examples

- Route planning (e.g., GPS)
- Robot navigation to a target
- AI coding agent (fix a bug)

Key idea

Choose actions that help achieve a goal, possibly by planning ahead.

4 Utility-Based Agent

Uses a utility function to handle trade-offs and preferences.



Examples

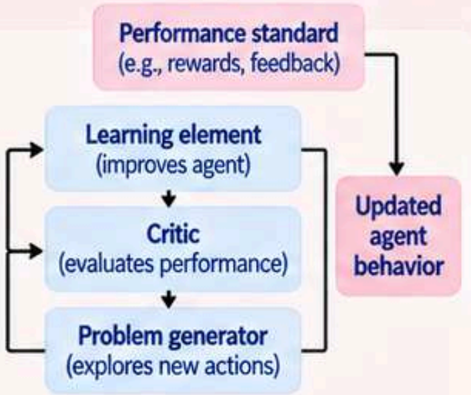
- Self-driving car (safety, time, comfort)
- Recommendation system (relevance vs. diversity)
- Resource management (cost vs. performance)

Key idea

Choose the action with the highest expected utility, balancing multiple objectives.

5 Learning Agent

Improves its behavior through experience.



Examples

- LLM-based agents that learn from user feedback
- Game-playing agents (e.g., AlphaGo)
- Robots that learn from trial and error

Key idea

Use experience and feedback to improve over time; can be combined with any of the above types.

The first four are basic agent-program families; learning can improve agents of these types.

In practice, modern AI systems often combine ideas from multiple types (e.g., goal-based + utility-based + learning).

Question 2: Consider the given scenario and demonstrate your understanding by answering the following sub-questions.

Hospital Delivery Robot

A hospital deploys an autonomous robot to deliver sealed medicine containers from its pharmacy to different wards. The robot receives a destination ward and a digital floor map at the beginning of each delivery. Its cameras, laser scanner, wheel encoders, and door-status receiver help it estimate its position and detect people, trolleys, and closed passages. Patients and staff move unpredictably, and some corridor sections cannot be observed until the robot approaches them.

The robot maintains an internal map of accessible corridors and updates it whenever it detects a temporary blockage. Unlike a machine that always follows one fixed route, it generates possible sequences of movements and checks whether each sequence can reach the assigned ward. It selects the first safe route found by its search procedure that satisfies the delivery goal. If a corridor becomes blocked, it searches again from its current estimated position. It does not rank successful routes using a numerical score for speed, energy use, or convenience.

The robot can move, turn, stop, request an automatic door to open, and notify ward staff when it arrives. Its planning procedure and action model are supplied by engineers and do not improve from previous deliveries. The hospital evaluates correct delivery and safe movement; the robot is not designed to reproduce human thought processes or human mannerisms.

a) Construct a PEAS description for the robot, including at least two suitable performance criteria.

b) Identify the most appropriate agent type and support your answer with evidence from the scenario. Explain how its use of future action sequences differs from a model-based reflex agent.

c) Classify the task environment as deterministic or stochastic, episodic or sequential, static or dynamic, and discrete or continuous. Justify each classification for the physical robot.

d) Select the most appropriate of the four AI approaches and justify your answer using the hospital's stated design objective.

e) Formulate this delivery task as a search problem by identifying the initial state, possible actions, goal test, and one reasonable path-cost measure. Explain how the proposed cost measure could improve route selection.

Question 3: Consider the given scenario and demonstrate your understanding by answering the following sub-questions.

Autonomous Agricultural Drone

A farm uses an autonomous drone to inspect crops and apply small amounts of water to dry sections of a field. The drone receives a field boundary map, its remaining water capacity, and a list of crop areas requiring inspection. Cameras, soil-moisture readings transmitted by ground sensors, GPS, and wind sensors provide observations. Some ground sensors report late, and changing wind can affect both travel time and the accuracy of water application.

The drone maintains estimates of crop condition, its position, battery charge, and remaining water. Before choosing an action, it considers several options, such as watering a nearby area, flying to a more severely affected area, collecting another observation, or returning to the charging station. It predicts possible outcomes and assigns each outcome a numerical value based on crop benefit, energy consumption, wasted water, and safety. It combines these values with estimated outcome probabilities and selects the permitted action with the highest expected score.

Safety limits prohibit flying outside the field and require enough estimated battery charge for a safe return. The scoring formula and probability model are fixed by the designers; the drone does not update them from previous missions. Its purpose is to make effective decisions under uncertainty rather than to imitate the behavior of a human farmer.

a) Identify all four PEAS components for the drone. Distinguish between a performance measure and an actuator using examples from the scenario.

b) Identify the most appropriate agent type. Explain why simply knowing whether an action achieves a goal is insufficient to describe this drone's decision process.

c) Classify its environment as deterministic or stochastic, episodic or sequential, static or dynamic, and discrete or continuous. Justify each answer.

d) Which of the four approaches to AI is most consistent with the stated design purpose? Explain your reasoning.

e) Explain the difference between maximizing expected utility and guaranteeing a successful outcome. Describe one situation in which collecting another sensor reading could be preferable to immediately watering a crop area.

Question 4: Consider the given scenario and demonstrate your understanding by answering the following sub-questions.

Adaptive Warehouse Robot

An online retailer uses an autonomous robot to move packages from storage shelves to dispatch stations. The warehouse contains other delivery robots and human workers. Cameras, laser scanners, wheel encoders, and barcode readers provide observations, but tall shelves hide parts of nearby aisles. Workers may move carts without warning, and congestion changes throughout the day. The robot controls its wheels, lifting platform, warning lights, and audible alert.

Initially, the robot uses an engineer-provided policy to choose movements and package-handling actions. During operation, it records delivery times, delays, near-collisions, and battery use. A feedback component evaluates completed deliveries using a reward that favors correct, timely delivery and penalizes unsafe behavior and wasted energy. An updating component uses this feedback to improve estimates of travel time and revise which permitted actions the robot tends to choose in similar situations.

A separate exploration component occasionally suggests a less-used aisle so that the robot can collect useful experience. Every proposed action must still satisfy fixed safety restrictions. Over several weeks, its route choices change as it learns where congestion is likely. The retailer judges the robot by delivery performance and safety, without requiring it to behave or reason like a human warehouse worker.

a) Develop a PEAS description for the robot, taking account of package handling and interaction with other warehouse users.

b) Identify the most appropriate agent type. Cite two details showing that its behavior changes through experience rather than only through updates to its current location.

c) Classify the environment as deterministic or stochastic, episodic or sequential, static or dynamic, and discrete or continuous. Justify each classification.

d) Choose the most appropriate of the four AI approaches. Also classify the environment as single-agent or multi-agent and explain how other decision-making agents affect the robot.

e) Identify the performance element, learning element, critic, and problem generator in this system. Explain one possible benefit and one possible risk of exploring a less-used aisle.

Question 5: Consider the given scenario and demonstrate your understanding by answering the following sub-questions.

Automated Parcel Inspection System

A distribution centre uses an automated inspection system to decide whether an individual parcel should be accepted, rejected, or sent for manual inspection. An operator places one parcel inside an enclosed inspection chamber. While the parcel is inspected, the conveyor remains stopped and the parcel cannot change position. A camera reads the label and checks the exterior, a scale measures weight, and a dimension sensor measures length, width, and height.

For this task, acceptance depends only on the readable label, visible exterior condition, weight, and dimensions. Assume that the sensors report these required properties completely and without error. Hidden contents are outside the task definition. The system applies fixed rules: a damaged exterior triggers rejection; an unreadable label triggers manual inspection; and a parcel exceeding a size or weight limit also triggers manual inspection. If none of these conditions applies, it accepts the parcel. The rules include a fixed priority order so that every combination of readings produces one decision.

The system then activates the appropriate sorting gate and displays the decision. It does not retain information about earlier parcels, predict future outcomes, or learn from operator feedback. Assume that each sorting action works exactly as commanded and that decisions about one parcel do not affect the acceptance criteria or available actions for another. The designers assess correct rule application and reliable sorting rather than similarity to human reasoning.

a) Identify the four PEAS components. Clearly separate the sensors that obtain information from the actuators that carry out the decision.

b) Identify the most appropriate agent type. Explain why the system does not require an internal model of previous parcels to perform the stated task.

c) Under the stated assumptions, classify the environment as deterministic or stochastic, episodic or sequential, and static or dynamic. Then discuss which aspects of the task are discrete and which measured quantities are continuous.

d) Choose the most appropriate of the four AI approaches and justify your answer. Would using fixed rules alone prove that the system thinks like a human? Explain.

e) Suppose the scale becomes noisy and the system must combine several past readings before deciding. Explain how this change affects observability and the need for internal state. Does storing readings alone make the system a learning agent? Justify your answer.